Cross-Paper Synthesis & Methodological Evolution

Deconstructing Exchange Rate Dynamics, Pass-Through Rigidities, and Deep Learning Architectures

COMPREHENSIVE ACADEMIC ANALYSIS • 4 RESEARCH PAPERS • SYSTEMATIC CONCEPTUAL MAPPING

This report provides an in-depth breakdown of four seminal papers spanning macroeconomic exchange rate pass-through, firm-level multilateral pricing behavior, high-frequency currency forecasting, and enterprise-level AI decision support. For each paper, the **Abstract**, **Methodology**, and **Conclusion** are rigorously delineated, accompanied by an explicit demonstration of how each paper connects internally into its logical next steps. Finally, a synthesized **Thought Process Narrative** demonstrates how the research community sequentially arrives from foundational macro observations to multi-layered, operational AI companions in global executive strategy.

1. Systematic Deconstruction of Individual Papers

Paper 1: Exchange Rate Pass-Through for European Union Countries

Authors: Duc Hong Vo & Nam Thanh Vu (2024) | Source: PLOS ONE, 19(8): e0309527

ABSTRACT

Exchange rate pass-through (ERPT) reflects the degree to which nominal exchange rate fluctuations transmit into domestic prices. Investigating 16 European Union member states from January 2006 to December 2022, the authors examine whether ERPT to import prices is linear or nonlinear, and how it evolves under extreme geopolitical shocks (the 2008 Great Recession, COVID-19 pandemic, and Russia-Ukraine war). The paper finds robust evidence for a linear, diminishing short-run pass-through (elasticity of 0.28) and zero long-run pass-through across the union, rejecting the strict Purchasing Power Parity (PPP) hypothesis in the short run. Furthermore, ERPT is heightened and more persistent in countries highly dependent on imports and during intervals of pronounced macroeconomic instability.

METHODOLOGY

The empirical strategy is anchored on the micro-foundations of exporter pricing behavior (Campa & Goldberg, 2005), specifying import prices ($mp_{i,t}$) as a function of trade-weighted nominal effective exchange rates ($er_{i,t}$), foreign exporter marginal costs ($w_{i,t}$ proxied by real exchange rates and domestic consumer price indices), domestic output gap ($y_{i,t}$ derived via the Hodrick-Prescott filter on industrial production), and competitor producer prices ($pp_{i,t}$). The authors deploy **Panel Autoregressive Distributed Lag (ARDL)** models using both Mean Group (MG) and Pooled Mean Group (PMG) estimators, validated by the Hausman test for long-run slope homogeneity. Non-linearities are examined using partial sum decompositions (er^{+} and er^{-}) within an asymmetric ARDL framework, supplemented by sub-sample rolling windows (2008–2012, 2013–2017, 2018–2022) and trade-openness segmentation.

CONCLUSION

The study concludes that ERPT across the EU is predominantly incomplete ($\beta = 0.28$ in the short run, implying a 10% currency appreciation yields a 2.8% import price drop), linear (symmetric between depreciations and appreciations), and subject to structural secular decline over decades. This confirms that exporters widely practice Local Currency Pricing (LCP). However, pass-through regimes become significantly more acute during crisis episodes (2018–2022 short-run ERPT jumped to ~0.49) and in economies with elevated import-to-GDP ratios, forcing central banks and policymakers to calibrate monetary interventions dynamically to external trade dependencies.

> ***Internal Logical Connection & Next Step:*** *Vo & Vu establish that aggregate border prices fail to adjust 1:1 with currencies due to incomplete pass-through and exporter markups. However, panel macro models treat nations as homogenous aggregates. The necessary logical question is: What happens at the micro-firm level inside these economies? Why do exporters refrain from passing exchange shocks into prices, and does an exchange shock in one destination bleed into non-shocked destinations? This necessitates moving to firm-level, micro-econometric transaction datasets.*

Paper 2: Exchange Rate Pass-Around

Authors: Matthieu Crozet, Julian Hinz, & Federico Trionfetti (2024) | Source: SSRN Working Paper No. 4776555

ABSTRACT

Challenging the conventional bilateral paradigm of exchange rate pass-through, this study introduces the phenomenon of the **Exchange Rate Pass-Around (ERPA)**. The authors argue that incomplete ERPT is not purely a function of variable markups or foreign marginal costs, but stems from the unwillingness or inability of certain exporters to engage in perfect third-degree price discrimination across markets (Pricing-to-Market, PTM). Instead, firms constrained by fixed market-segmentation costs adopt a "single-price strategy," setting a uniform Free-On-Board (FOB) price across multiple destinations. Exploiting the sudden, massive appreciation of the Swiss Franc (CHF) in January 2015, the authors prove that French exporters selling in Euros to both Switzerland and the Eurozone raised their prices not only in Switzerland (incomplete ERPT) but also across uninvolved Eurozone markets (ERPA).

METHODOLOGY

The paper develops a multi-country monopolistic competition framework with destination-specific marginal costs (c_j) and an endogenous fixed cost (F_t) required to implement bilateral pricing-to-market. The empirical verification exploits French customs declarations from January 2014 to December 2015 encompassing over 6 million transaction records. The empirical identification leverages two Difference-in-Differences (DiD) specifications: (1) a two-way fixed effects (TWFE) model with firm-product-destination and firm-product-time fixed effects evaluating Swiss price divergence against European controls; and (2) a DiD model identifying ERPA by contrasting exposed Euro-invoicing exporters to Switzerland against non-exposed Eurozone exporters. Firm price-discrimination capability is empirically isolated using Theil entropy indices decomposed into within-market (***TheilW***) and between-market (***TheilB***) dispersions.

CONCLUSION

The findings confirm that French exporters of differentiated products exhibited an incomplete pass-through of 78.6%–86.4% on the Swiss market while concurrently raising their Eurozone export prices by 1.51% (ERPA). Crucially, the ERPA effect is concentrated entirely among low-***TheilB*** firms (those lacking price discrimination infrastructure) and is virtually zero for high-***TheilB***, large multinational firms. Moreover, ERPA magnitude is amplified by geographic proximity to Switzerland and high initial Swiss-to-Eurozone price premia, but attenuates over time as firms gradually incur the fixed costs to switch from single-pricing to PTM.

> ***Internal Logical Connection & Next Step:*** *Crozet et al. reveal that firm-level pricing is deeply entangled across borders: when currency shocks hit, firms cannot instantly adjust prices without creating spillover shocks in other regions or destroying demand. To prevent disastrous pricing moves, corporate executives must anticipate exchange rate shifts before they occur. However, traditional econometric models fail to capture high-frequency, non-linear FX volatility, demanding deep learning architectures like LSTMs designed specifically for sequential, non-stationary financial data.*

Paper 3: Foreign Exchange Rate Prediction Using Long-Short Term Memory: A Case Study in COVID-19 Pandemic

Authors: Hasna Haifa Zahrah, Siti Sa'adah, & Rita Rismala (2020) | Source: International Journal on ICT, 6(2): 94–105

ABSTRACT

Forecasting foreign exchange rates is notoriously difficult due to extreme volatility, market psychology, and macroeconomic noise, all of which were exacerbated by the unprecedented disruption of the COVID-19 pandemic. This study implements a Long Short-Term Memory (LSTM) recurrent neural network to forecast the EUR/USD exchange rate across both 1-hour and daily timeframes. The authors systematically calibrate architectural hyperparameters (number of hidden layers, neuron density, and dropout regularization) using Adam optimization and Min-Max scaling. Testing the calibrated model on 2020 crisis data and comparing performance against stable pre-pandemic years (2018 and 2019), the paper demonstrates that LSTMs maintain robust predictive capability during major exogenous crises, achieving a minimal test Root Mean Square Error (RMSE) of 0.00135.

METHODOLOGY

Historical EUR/USD trading data spanning May 4, 2010 to October 19, 2020 were acquired from MetaQuotes MT4, encompassing 2,718 daily records and 64,897 hourly records. The pipeline utilizes Min-Max normalization mapping close prices to [0, 1]. Model selection evaluates grid candidates: hidden layers [1, 2, 3], neurons per layer [1, 5, 10], and dropout rates [No dropout, 0.1]. Optimization is conducted using Adam with Mean Squared Error (MSE) loss and early stopping callbacks (patience = 30 epochs, maximum 1,000 epochs). Holdout validation allocates 67% of pre-2020 data to training and 33% to validation, reserving 2020 as the out-of-sample test horizon. Performance is evaluated via RMSE against actual closing prices.

CONCLUSION

The 1-hour timeframe achieved superior predictive accuracy (***RMSE = 0.00135*** with 1 layer, 5 neurons, no dropout) compared to the daily timeframe (***RMSE = 0.00624*** with 2 layers, 10 neurons, 0.1 dropout), primarily driven by the massive volume of high-frequency training sequences (59,914 records) enabling the gates to learn fine-grained intraday dynamics. While pandemic turmoil in early 2020 (sharp Federal Reserve rate cut bets followed by ECB stimulus announcements) caused a temporary spike in prediction error relative to the steady market conditions of 2019 (***RMSE = 0.00111***), the LSTM successfully tracked the non-linear trend and recovered tight tracking during subsequent sideways and trending regimes.

> ***Internal Logical Connection & Next Step:*** *Zahrah et al. prove that LSTM networks can accurately forecast EUR/USD time series even across violent macroeconomic regime shifts. Yet, predicting raw exchange rates is purely a financial trading tool. In a global enterprise, an exchange rate prediction is useless in isolation—it must be connected to real-world corporate operations (bookings, daily active users, demand elasticities, and profit margins) to guide board-level pricing decisions under foreign exchange stress.*

Paper 4: AI as a Decision Companion: Supporting Executive Pricing and FX Decisions in Global Enterprises Through LSTM Forecasting

Authors: Wesley Leeroy & Gordon C. Leeroy (2025) | Source: Journal of Risk and Financial Management, 18(10): 542

ABSTRACT

Bridging the divide between operational AI and board-level strategic choices, this paper investigates how deep learning can serve as an executive "decision companion" when global enterprises confront severe macroeconomic shocks. Grounded in Herbert Simon’s bounded rationality, the study uses multinational digital gaming platform Roblox Corporation as an empirical case study. Roblox faces acute currency

exposure because it derives over 75% of Daily Active Users (DAU) internationally and sells virtual currency (Robux) across volatile currencies but reports consolidated earnings in USD. Developing a multi-feature LSTM architecture, the authors forecast quarterly bookings and simulate counterfactual scenarios: a 10% euro depreciation and an executive intervention raising European prices by 10%. The findings prove that currency depreciation slashes bookings and profits by ~6%, and raising European prices completely fails to restore profits due to regional demand elasticity.

METHODOLOGY

The methodology combines deep sequential time-series forecasting with Agent-Based Modeling (ABM) principles and SHAP interpretability. Quarterly enterprise data (2010–2025) integrates 16 input features: firm-specific fundamentals (bookings, DAU, engagement hours, ABPDAU, regional revenue shares), macroeconomic drivers (USD/EUR, USD/GBP, USD/BRL, broad USD index from FRED), and strategic levers (regional price multipliers, viral and policy shocks). The neural architecture comprises an input layer (lookback window of 6 quarters × 16 features), a 64-unit LSTM layer, a 32-unit Dense layer with ReLU activation, and a single scalar dense output trained under MAE with the Adam optimizer. Recursive 4-quarter out-of-sample forecasting is coupled with regional attribution analysis decomposing consolidated impacts across EU, US, LATAM, and APAC agents.

CONCLUSION

The empirical results confirm three core hypotheses: (H1) A 10% euro depreciation cuts quarterly bookings from 1,127 to 1,056 units (−6.3%) and operating profits from $338M to $317M; (H2) Intervening by raising European prices by 10% leaves bookings and profits stagnant at 1,057 units and $317M, because European price elasticity causes demand to collapse, offsetting higher unit revenues; and (H3) 74% of the losses are concentrated directly in Europe (−42.6 units), with minimal spillovers to the US (−8.6 units) and LATAM/APAC (−2.8 units each). The paper concludes that AI acts not as an autonomous decision-maker, but as a crucial companion that protects C-suite executives from naïve pricing heuristics.

> ***Internal Logical Connection & Synthesis:*** *Leeroy & Leeroy represent the operational zenith of the lineage: they integrate the macroeconomic price rigidities of Vo & Vu, the cross-market pricing constraints of Crozet et al., and the recurrent deep learning capabilities of Zahrah et al. into a unified executive management system.*

2. Cross-Paper Synthesis: The Unified Conceptual Thread

The four papers are not isolated investigations; they represent four progressive resolutions of the exact same underlying economic tension: **how nominal exchange rate shocks interact with enterprise pricing structures, consumer demand elasticities, and decision-making uncertainty.**

Dimension	Paper 1 (Vo & Vu)	Paper 2 (Crozet et al.)	Paper 3 (Zahrah et al.)	Paper 4 (Leeroy & Leeroy)
Level of Analysis	Macroeconomic / Cross-Country Panel (16 EU states)	Microeconomic / Firm-level Exporters (French Customs)	Financial Asset Market / High-Frequency Time Series	Enterprise Strategy / Multinational Platform (Roblox)
Primary Methodology	Panel ARDL (PMG / MG Estimators) & Cointegration	Two-Way Fixed Effects & Event Study DiD	Univariate Recurrent Neural Network (LSTM)	Multivariate LSTM + Scenario Simulation + SHAP
Core Theoretical Concept				

Dimension	Paper 1 (Vo & Vu)	Paper 2 (Crozet et al.)	Paper 3 (Zahrah et al.)	Paper 4 (Leeroy & Leeroy)
	Exchange Rate Pass-Through (ERPT) & PPP Breakdown	Exchange Rate Pass-Around (ERPA) & Single-Pricing	Nonlinear Sequential Learning & Memory Gates	Bounded Rationality & Executive Decision Companion
Key Finding / Mechanism	Incomplete ERPT (0.28); heightened pass-through during crises	Firms unable to price-discriminate spill price shocks to 3rd markets	LSTM captures violent EUR/USD crisis shifts (RMSE: 0.00135)	Raising local prices under FX shocks fails due to demand elasticity

3. Step-by-Step Thought Process: Arriving from One Paper to the Next

The intellectual journey bridging these papers follows a deliberate, inductive-to-deductive continuum that mirrors the historical evolution of modern international corporate finance and artificial intelligence:

Step 1 **From Macro Disconnect to Micro Friction: Paper 1 → Paper 2**

The Conceptual Dilemma: Vo & Vu document empirically that at the aggregate border level across 16 European nations, exchange rate pass-through is incomplete ($\beta = 0.28$) and linear. When currencies swing, import prices barely move by a third of the shock. But macro panels are blind to the behavioral mechanics inside the exporting firm. *Why do exporters fail to pass currency depreciation through to prices?* Traditional theory blamed currency invoicing (LCP vs. PCP) or variable markups, but assumed markets were treated independently in bilateral pairs.

The Theoretical Leap to Paper 2: Crozet, Hinz, and Trionfetti recognize that firms do not operate in bilateral vacuums. In the real world, segmenting markets into customized pricing-to-market tiers requires massive managerial and organizational fixed costs (F_t). Small and medium exporters adopt a *single-price strategy*. The authors demonstrate that incomplete pass-through on the shocked market (Switzerland) is the mathematical consequence of firms adjusting a single common price, which involuntarily spills price increases into completely unaffected markets (the Eurozone). Paper 2 thereby provides the micro-structural behavioral foundation for the incomplete pass-through observed at the macro level in Paper 1.

Step 2 **From Pricing Entanglement to Predictive Foresight: Paper 2 → Paper 3**

The Operational Dilemma: Crozet et al. prove that multinational price setting is a high-stakes balancing act: an exporter cannot simply raise prices in response to an adverse currency shock without triggering unintended horizontal price repercussions across all export destinations. If pricing changes are sticky, costly, and dangerous, corporate treasuries and trading desks cannot afford to act purely reactively after exchange rates collapse. They require predictive models capable of anticipating exchange rate movements across volatile, non-linear market environments.

The Technological Leap to Paper 3: Classical econometric forecasting (ARIMA, structural VAR) relies on linear assumptions and Gaussian stationarity, which notoriously fail during black-swan shocks. Zahrah, Sa'adah, and Rismala pivot to deep learning, demonstrating that Long Short-Term Memory (LSTM) neural networks—equipped with input, forget, and output gating mechanisms—can overcome the vanishing gradient problem, learn long-range temporal dependencies, and accurately forecast the EUR/USD exchange rate through the violent regime shifts of the COVID-19 pandemic.

Step 3 **From Pure Price Forecasting to Enterprise Strategic Simulation: Paper 3 → Paper 4**

The Strategic Dilemma: While Zahrah et al. validate the technical efficacy of LSTM networks, their model remains confined to a standalone financial trading environment (predicting close prices on MT4). In an enterprise boardroom, knowing that EUR/USD will depreciate by 10% is incomplete information. The executive needs to know: *If the Euro depreciates, what will happen to consolidated revenue, daily user engagement, and net operating margin? And if we counteract by raising European prices by 10%, will that recover our profits?*

The Climax in Paper 4: Wesley Leeroy & Gordon Leeroy synthesize the entire research continuum. They take the LSTM deep learning architecture validated by Zahrah et al., expand it into a multivariate enterprise forecasting engine (16 features combining firm metrics and FRED macro data), and use it to model the pricing-demand trade-offs identified by Vo & Vu and Crozet et al. By running counterfactual scenario simulations on Roblox, they prove that executive heuristics (e.g., "raise prices to protect dollar margins") trigger catastrophic demand contractions due to regional price elasticity. AI thus becomes the ultimate *decision companion*, resolving bounded rationality and providing C-suite executives with a quantitative sandbox to simulate trade-offs before executing fatal pricing blunders.